

2.10 stdarg.h

The stdarg header defines several macros used to get the arguments in a function when the number of arguments is not known.

Macros:

```
va_start();  
va_arg();  
va_end();
```

Variables:

```
typedef va_list
```

2.10.1 Variables and Definitions

The `va_list` type is a type suitable for use in accessing the arguments of a function with the stdarg macros.

A function of variable arguments is defined with the ellipsis (`, ...`) at the end of the parameter list.

2.10.2 `va_start`

Declaration:

```
void va_start(va_list ap, last_arg);
```

Initializes *ap* for use with the `va_arg` and `va_end` macros. *last_arg* is the last known fixed argument being passed to the function (the argument before the ellipsis).

Note that `va_start` must be called before using `va_arg` and `va_end`.

2.10.3 `va_arg`

Declaration:

```
type va_arg(va_list ap, type);
```

Expands to the next argument in the parameter list of the function with type *type*. Note that *ap* must be initialized with `va_start`. If there is no next argument, then the result is undefined.

2.10.4 `va_end`

Declaration:

```
void va_end(va_list ap);
```

Allows a function with variable arguments which used the `va_start` macro to return. If `va_end` is not called before returning from the function, the result is undefined. The variable argument list *ap* may no longer be used after a call to `va_end` without a call to `va_start`.

Example:

```
#include<stdarg.h>
#include<stdio.h>

void sum(char *, int, ...);

int main(void)
{
    sum("The sum of 10+15+13 is %d.\n",3,10,15,13);
    return 0;
}

void sum(char *string, int num_args, ...)
{
    int sum=0;
    va_list ap;
    int loop;

    va_start(ap,num_args);
    for(loop=0;loop<num_args;loop++)
        sum+=va_arg(ap,int);

    printf(string,sum);
    va_end(ap);
}
```